Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

ACID PROPERTIES & ISOLATION

2520030078

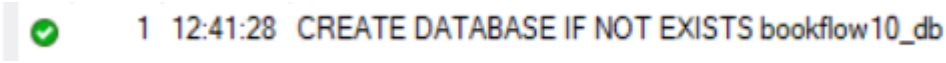
M.Apoorva

Section – 3

Create database:

```
CREATE DATABASE IF NOT EXISTS bookflow10_db;
```

Output:



Use database:

```
USE bookflow10_db;
```

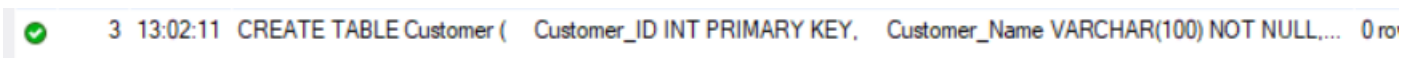
Output:



Create Customer Table:

```
CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15),
    City VARCHAR(50)
);
```

Output:



Create Account Table:

```sql
CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2),
    Branch VARCHAR(50),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
```

Output:



Create Transaction Table:

```sql
CREATE TABLE Bank_Transaction (
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (Account_No)
    REFERENCES Account(Account_No)
);
```

Output:



Insert Customer Data:

```
INSERT INTO Customer
(Customer_ID, Customer_Name, Phone, City)
VALUES
(101, 'Sushma', '9876543210', 'Hyderabad'),
(102, 'Sush', '9876543211', 'Vijayawada'),
(103, 'Sushuu', '9876543212', 'Bangalore'),
(104, 'Chuchuma', '9876543213', 'Chennai'),
(105, 'Sonu', '9876543214', 'Hyderabad');
```

Output:

```
6  13:05:54  INSERT INTO Customer (Customer_ID, Customer_Name, Phone, City) VALUES (101, 'Sushma', '9876543210', 'Hy...
```

Insert Account Data:

```
INSERT INTO Account
(Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001, 101, 'Savings', 50000, 'Hyderabad'),
(10002, 102, 'Savings', 75000, 'Vijayawada'),
(10003, 103, 'Current', 120000, 'Bangalore'),
(10004, 104, 'Savings', 45000, 'Chennai'),
(10005, 105, 'Current', 90000, 'Hyderabad');
```

Output:

```
8  13:26:44  INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Savi...  5 row(
```

Insert Bank Transaction Data:

```
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),
(10002, 'DEPOSIT', 15000),
(10003, 'WITHDRAW', 20000),
(10004, 'DEPOSIT', 5000),
(10005, 'WITHDRAW', 10000);
```

Output:

```
9  13:27:56  INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000),...  5 r
```

Display Data:

```
SELECT * FROM Customer;
```

Output:

Customer_ID	Customer_Name	Phone	City
101	Sushma	9876543210	Hyderabad
102	Sush	9876543211	Vijayawada
103	Sushuu	9876543212	Bangalore
104	Chuchuma	9876543213	Chennai
105	Sonu	9876543214	Hyderabad
NULL	NULL	NULL	NULL

```
SELECT * FROM Account;
```

Output:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

```
SELECT * FROM Bank_Transaction;
```

Output:

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-16 13:27:56
2	10002	DEPOSIT	15000.00	2026-08-16 13:27:56
3	10003	WITHDRAW	20000.00	2026-08-16 13:27:56
4	10004	DEPOSIT	5000.00	2026-08-16 13:27:56
5	10005	WITHDRAW	10000.00	2026-08-16 13:27:56
NULL	NULL	NULL	NULL	NULL

Start A Transaction:

```
START TRANSACTION;
```

Output:

```
13  13:31:49  START TRANSACTION
```

Deposit Money:

```
UPDATE Account
SET Balance = Balance + 5000
WHERE Account No = 10001;
```

Output:

```
14  13:33:05  UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001
```

Check Balance:

```
SELECT *
FROM Account
WHERE Account_No = 10001;
```

Output:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Commit Transaction:

```
COMMIT;
```

Output:

```
16  13:34:35  COMMIT
```

Withdraw Money:

```
UPDATE Account
SET Balance = Balance - 10000
WHERE Account_No = 10001;
```

Output:

```
17  13:35:53  UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001
```

Check Balance:

```
SELECT *
FROM Account
WHERE Account_No = 10001;
```

Output:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	45000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Rollback:

```
ROLLBACK;
```

Output:

19 13:45:18 ROLLBACK

Start Transaction:

```
START TRANSACTION;
```

Output:

13 13:31:49 START TRANSACTION

First Update:

```
UPDATE Account
SET Balance = Balance + 5000
WHERE Account_No = 10001;
```

Output:

22 13:51:41 UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001

Create Savepoint:

```
SAVEPOINT Deposit1;
```

Output:

23 13:52:33 SAVEPOINT Deposit1

Second Update:

```
UPDATE Account
SET Balance = Balance - 3000
WHERE Account_No = 10002;
```

Output:

```
24  14:00:28  UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002
```

Create Second Savepoint:

```
SAVEPOINT Withdrawal1;
```

Output:

```
25  14:02:17  SAVEPOINT Withdrawal1
```

Third Update:

```
UPDATE Account
SET Balance = Balance + 10000
WHERE Account_No = 10003;
```

Output:

```
26  14:03:52  UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10003
```

Rollback To Savepoint:

```
ROLLBACK TO SAVEPOINT Withdrawal1;
```

Commit:

```
COMMIT;
```

Output:

```
30  14:07:18  COMMIT
```

Check Accounts:

```
SELECT *
FROM Account;
```

Output:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	72000.00	Vijayawada
10003	103	Current	130000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL